Installation de Redis Cluster avec keepalived &Ha proxy sur 2VM

Répertoires de fichiers VM Master : Ne pas prendre en compte start-redis2.sh

```
root@honor13:/home/redis/redis-server# ls -l
total 36
-rw-r--r-- 1 root root  378 May 19 12:47 check_haproxy.sh
-rw-r--r-- 1 root root 2886 May 19 08:13 docker-compose.yml
-rw-r--r-- 1 root root  578 May 19 12:53 dockerfile
-rw-r--r-- 1 root root 1215 May 19 12:37 haproxy.cfg
-rw-r--r-- 1 root root  602 May 19 12:52 keepalived.conf
-rw-r--r-- 1 root root  685 May 16 16:39 redis.conf
-rwxr-xr-x 1 root root  247 May 16 06:30 start-redis2.sh
-rw-r--r-- 1 root root  297 May 19 12:38 start.sh
root@honor13:/home/redis/redis-server#
```

Configuration du proxy TCP HA Proxy, utilisé pour équilibrer la charge entre les instances Redis du cluster.

```
  GNU nano 7.2                    haproxy.cfg
global
    log /dev/log local0
    log /dev/log local1 notice
    log stdout format raw local0
    maxconn 4096

defaults
    log global
    mode tcp
    option tcplog
    option redispatch
    timeout connect 5s
    timeout client 30s
    timeout server 30s
    retries 3

listen stats
    bind *:8081
    mode http
    stats enable
    stats uri /
    stats refresh 10s
    stats admin if TRUE
    stats show-node

frontend redis_front
    bind *:6379
    mode tcp
    default_backend redis_cluster

backend redis_cluster
    mode tcp
    balance roundrobin

    # Specific configuration for Redis Cluster
    tcp-check send PING\r\n
    tcp-check expect string +PONG
    tcp-check expect string cluster_enabled:1
    tcp-check send QUIT\r\n
    tcp-check expect string +OK

    # Redis server cluster
    server redis1 10.0.50.149:6379 check inter 1s fall 2 rise 1
```

Script de démarrage principal du conteneur HAProxy/Keepalived

```
  GNU nano 7.2                                  start.sh
#!/bin/sh

# Verify HAProxy configuration
if ! haproxy -f /etc/haproxy/haproxy.cfg -c; then
    echo "ERROR: Invalid HAProxy configuration"
    exit 1
fi

# Start HAProxy in daemon mode
haproxy -f /etc/haproxy/haproxy.cfg -D

# Start Keepalived (main process)
exec /usr/sbin/keepalived -n -l "$@"
```

Script de supervision utilisé par Keepalived :

```
  GNU nano 7.2                    check_haproxy.sh
#!/bin/sh

# Define the port to check (e.g., HAProxy frontend port)
PORT=8081

# Check if HAProxy is running
if ! pidof haproxy > /dev/null; then
    echo "HAProxy is not running"
    exit 1
fi

# Check if proxy is listening on the expected port
if ! netstat -tln | grep -q ":${PORT}"; then
    echo "HAProxy is not listening on port ${PORT}"
    exit 2
fi

# All checks passed
```

Configuration de Keepalived pour assurer la **haute disponibilité (HA)** :

Définissons une instance VRRP (vrrp_instance) en « état MASTER » sur l'interface réseau eth0

* Associons un IP virtuel (VIP) partagé entre la VM Master et Backup.

* Utilisons check_haproxy.sh comme script de vérification.

```
  GNU nano 7.2                          keepalived.conf *
global_defs {
    script_user keepalived_script
    enable_script_security
}

vrrp_script check_haproxy {
    script "/etc/keepalived/check_haproxy.sh"
    interval 2
    fall 3
    rise 2
}

vrrp_instance VI_1 {
    state MASTER
    interface eth0
    virtual_router_id 51
    priority 100
    advert_int 1
    authentication {
        auth_type PASS
        auth_pass redis-ha  #change password
    }
    virtual_ipaddress {
        10.0.50.4/24   #replace with  VIP available
    }
    unicast_peer {
        10.0.50.103  #@ip HAProxy backup
    }
    track_script {
        check_haproxy
    }
}
```

Dockerfile

```
  GNU nano 7.2                          dockerfile
FROM instantlinux/haproxy-keepalived

RUN apk update && apk add procps iproute2

# Add a non-root user for script execution
RUN id keepalived_script || adduser -D -s /bin/sh keepalived_script

# Copy configuration file and scripts into the image
COPY keepalived.conf /etc/keepalived/keepalived.conf
COPY check_haproxy.sh /etc/keepalived/check_haproxy.sh
COPY start.sh /start.sh

# Set proper permissions for the scripts
RUN chown keepalived_script:keepalived_script /etc/keepalived/check_haproxy.sh && \
    chmod +x /etc/keepalived/check_haproxy.sh && \
    chmod +x /start.sh
```

Docker -compose yml

```
  GNU nano 7.2                    docker-compose.yml
version: "3.9"

services:
  redis-1:
    image: redis:latest
    container_name: redis1
    command: redis-server /usr/local/etc/redis/redis.conf
    volumes:
      - redis-data-1:/data
      - ./redis.conf:/usr/local/etc/redis/redis.conf
    ports:
      - "6379:6379"
    networks:
      - redis-cluster-net

  redis-2:
    image: redis:latest
    container_name: redis2
    command: redis-server /usr/local/etc/redis/redis.conf
    volumes:
      - redis-data-2:/data
      - ./redis.conf:/usr/local/etc/redis/redis.conf
    ports:
      - "6380:6379"
    networks:
      - redis-cluster-net

  redis-3:
    image: redis:latest
    container_name: redis3
    command: redis-server /usr/local/etc/redis/redis.conf
    volumes:
      - redis-data-3:/data
      - ./redis.conf:/usr/local/etc/redis/redis.conf
    ports:
      - "6381:6379"
    networks:
      - redis-cluster-net

  redis-replica-1:
    image: redis:latest
    container_name: redis-replica-1
    command: redis-server /usr/local/etc/redis/redis.conf

^G Help      ^O Write Out  ^W Where Is   ^K Cut       ^T Execute   ^C Location
```

2. Préparation des fichiers de configuration (VM Backup)

Répertoire de fichiers sur VM backup

```
-rwxr-xr-x 1 root root  385 May 19 13:09 check_haproxy.sh
-rw-r--r-- 1 root root  511 May 19 13:14 docker-compose.yml
-rw-r--r-- 1 root root  578 May 19 13:19 dockerfile
-rw-r--r-- 1 root root 1214 May 19 13:06 haproxy.cfg
-rw-r--r-- 1 root root  597 May 19 13:11 keepalived.conf
-rw-r--r-- 1 root root  297 May 19 13:08 start.sh
root@Redis0:~/redis-ha#
```

haproxy.cfg

```
Non sécurisé  https://173.240.15.216:8006/?console=lxc&xtermjs=1&vmid=1204&vmname=1204&node=node-20...

  GNU nano 7.2                          haproxy.cfg
global
    log /dev/log local0
    log /dev/log local1 notice
    log stdout format raw local0
    maxconn 4096

defaults
    log global
    mode tcp
    option tcplog
    option redispatch
    timeout connect 5s
    timeout client 30s
    timeout server 30s
    retries 3

listen stats
    bind *:8081
    mode http
    stats enable
    stats uri /
    stats refresh 10s
    stats admin if TRUE
    stats show-node

frontend redis_front
    bind *:6379
    mode tcp
    default_backend redis_cluster

backend redis_cluster
    mode tcp
    balance roundrobin

    # Specific configuration for Redis Cluster
    option tcp-check
    tcp-check send PING\r\n
    tcp-check expect string +PONG
    tcp-check send INFO\ cluster\r\n
    tcp-check expect string cluster_enabled:1
    tcp-check send QUIT\r\n

    # Redis server cluster
                                  [ Read 49 lines ]
```

start.sh

```
  GNU nano 7.2                                    start.sh *
#!/bin/sh

# Verify HAProxy configuration
if ! haproxy -f /etc/haproxy/haproxy.cfg -c; then
    echo "ERROR: Invalid HAProxy configuration"
    exit 1
fi

# Start HAProxy in daemon mode
haproxy -f /etc/haproxy/haproxy.cfg -D

# Start Keepalived (main process)
exec /usr/sbin/keepalived -n -l "$@"
```

check-haproxy.sh

```
  GNU nano 7.2                    check-haproxy.sh
#!/bin/sh

# Define the port to check (e.g., HAProxy frontend port)
PORT=8080

# Check if HAProxy is running
if ! pidof haproxy > /dev/null; then
    echo "HAProxy is not running"
    exit 1
fi

# Check if proxy is listening on the expected port
if ! netstat -tln | grep -q ":${PORT}"; then
    echo "HAProxy is not listening on port ${PORT}"
    exit 2
fi

# All checks passed
exit 0
```

keepalived.conf

```
  GNU nano 7.2                          keepalived.conf *
global_defs {
    script_user keepalived_script
    enable_script_security
}

vrrp_script check_haproxy {
    script "/etc/keepalived/check_haproxy.sh"
    interval 2
    fall 3
    rise 2
}

vrrp_instance VI_1 {
    state BACKUP
    interface eth0
    virtual_router_id 51
    priority 50
    advert_int 1
    authentication {
        auth_type PASS
        auth_pass redis-ha  #change password
    }
    virtual_ipaddress {
        10.0.50.4   #replace with  VIP available
    }
    unicast_peer {
        10.0.50.149  #@ip HAProxy master
    }
    track_script {
        check_haproxy
```

docker-compose.yml

```
  GNU nano 7.2                    docker-compose.yml
services:
  # Load Balancer + HA (Backup)
  lb:
    build:
      context: .
      dockerfile: Dockerfile
    image: haproxy-keepalived
    container_name: haproxy-backup
    volumes:
      - ./haproxy.cfg:/etc/haproxy/haproxy.cfg:ro
      - haproxy-keepalived-data:/var/lib/haproxy
    network_mode: host
    cap_add:
      - NET_ADMIN
      - NET_BROADCAST
      - NET_RAW
      - SYS_NICE
      - SYS_RESOURCE
    restart: unless-stopped

volumes:
  haproxy-keepalived-data:
    name: haproxy-keepalived-data
```

dockerfile

```
  GNU nano 7.2                    dockerfile
FROM instantlinux/haproxy-keepalived

RUN apk update && apk add procps iproute2

# Add a non-root user for script execution
RUN id keepalived_script || adduser -D -s /bin/sh keepalived_script

# Copy configuration file and scripts into the image
COPY keepalived.conf /etc/keepalived/keepalived.conf
COPY check_haproxy.sh /etc/keepalived/check_haproxy.sh
COPY start.sh /start.sh

# Set proper permissions for the scripts
RUN chown keepalived_script:keepalived_script /etc/keepalived/check_haproxy.sh && \
    chmod +x /etc/keepalived/check_haproxy.sh && \
    chmod +x /start.sh
```

Démarrage des conteneurs

docker-compose build VM MASTER

✅ Résultat sur la VM Master

```
root@honor13:/home/redis/redis-server# docker-compose build

redis-1 uses an image, skipping
redis-2 uses an image, skipping
redis-3 uses an image, skipping
redis-replica-1 uses an image, skipping
redis-replica-2 uses an image, skipping
redis-replica-3 uses an image, skipping
redis-cluster-setup uses an image, skipping
Building lb
[+] Building 19.4s (12/12) FINISHED                                  docker:defaul
 => [internal] load build definition from dockerfile                          0.9
 => => transferring dockerfile: 617B                                          0.3
 => [internal] load metadata for docker.io/instantlinux/haproxy-keepalived:1 13.3
 => [internal] load .dockerignore                                             0.5
 => => transferring context: 2B                                               0.0
 => [1/7] FROM docker.io/instantlinux/haproxy-keepalived:latest@sha256:6f33a7 0.0
 => [internal] load build context                                             0.7
 => => transferring context: 101B                                             0.0
 => CACHED [2/7] RUN apk update && apk add procps iproute2                    0.0
 => CACHED [3/7] RUN id keepalived_script || adduser -D -s /bin/sh keepalived 0.0
 => CACHED [4/7] COPY keepalived.conf /etc/keepalived/keepalived.conf         0.0
 => CACHED [5/7] COPY check_haproxy.sh /etc/keepalived/check_haproxy.sh       0.0
 => CACHED [6/7] COPY start.sh /start.sh                                      0.0
 => CACHED [7/7] RUN chown keepalived_script:keepalived_script /etc/keepalive 0.0
 => exporting to image                                                        0.3
 => => exporting layers                                                       0.0
 => => writing image sha256:f9ea492046bcbbde1ed75779d67547edba073e91893ffb0e6 0.1
 => => naming to docker.io/library/haproxy-keepalived                         0.1
```

VM BACKUP

```
root@Redis0:~/redis-ha# docker-compose build
Building lb
[+] Building 982.6s (12/12) FINISHED                                 docker:default
 => [internal] load build definition from dockerfile                          0.8s
 => => transferring dockerfile: 617B                                          0.0s
 => [internal] load metadata for docker.io/instantlinux/haproxy-keepalived:la 5.7s
 => [internal] load .dockerignore                                             0.6s
 => => transferring context: 2B                                               0.0s
 => [1/7] FROM docker.io/instantlinux/haproxy-keepalived:latest@sha256:6f33a 21.8s
 => => resolve docker.io/instantlinux/haproxy-keepalived:latest@sha256:6f33a7 0.9s
 => => sha256:6f33a711a079a7136e18a330c56ada23c47ce4846e2235c 1.29kB / 1.29kB 0.0s
 => => sha256:324726bd800ef2e575337047cc7ac3622c84bbf61968900 1.87kB / 1.87kB 0.0s
 => => sha256:a78710a286a221c232a96aad9295950ddb4fcabb5c7f58f 7.95kB / 7.95kB 0.0s
 => => sha256:f18232174bc91741fdf3da96d85011092101a032a93a388 3.64MB / 3.64MB 1.1s
 => => sha256:90a6ea9b350c20a5aeda0baa2a9163c0884cd774ee0 294.90kB / 294.90kB 1.3s
 => => sha256:b3e1c337128fab5fbbcc92408f4701ddfc53e10367804223869 964B / 964B 1.8s
 => => extracting sha256:f18232174bc91741fdf3da96d85011092101a032a93a388b79e9 0.5s
 => => sha256:297fed5149a654c878e54d6ea5453097faab691dc986b17c564 444B / 444B 2.6s
 => => sha256:4f4fb700ef54461cfa02571ae0db9a0dc1e0cdb5577484a6d75e6 32B / 32B 2.8s
 => => sha256:365ace6e88a505182333a0507e7a785868dac05d132ae 10.17MB / 10.17MB 4.3s
 => => extracting sha256:90a6ea9b350c20a5aeda0baa2a9163c0884cd774ee0590b19ac2 1.1s
 => => sha256:73db0b274f2ad267d254fce2b715d96e8efa011fb3fb1b8 1.01kB / 1.01kB 4.9s
 => => extracting sha256:b3e1c337128fab5fbbcc92408f4701ddfc53e10367804223869a 0.0s
 => => sha256:9cf3762ecb2f2881bb6a553786c9fd505c9aa09a1ba7fa7 1.09MB / 1.09MB 5.2s
 => => extracting sha256:365ace6e88a505182333a0507e7a785868dac05d132ae95bf0b1 0.8s
 => => extracting sha256:297fed5149a654c878e54d6ea5453097faab691dc986b17c5641 0.0s
 => => extracting sha256:4f4fb700ef54461cfa02571ae0db9a0dc1e0cdb5577484a6d75e 0.0s
 => => extracting sha256:9cf3762ecb2f2881bb6a553786c9fd505c9aa09a1ba7fa70e274 0.4s
 => => extracting sha256:73db0b274f2ad267d254fce2b715d96e8efa011fb3fb1b8ff89c 0.0s
```

docker compose up -d

```
root@honor13:/home/redis/redis-server# docker compose up -d
WARN[0000] Found orphan containers ([keepalived]) for this project. If you removed o
r renamed this service in your compose file, you can run this command with the --rem
ove-orphans flag to clean it up.
[+] Running 8/8
 ✓ Container redis3                Started                                    181.1s
 ✓ Container redis-replica-1       Started                                    182.2s
 ✓ Container redis-replica-2       Started                                    183.5s
 ✓ Container redis-replica-3       Started                                    184.6s
 ✓ Container haproxy-master        Started                                    184.3s
 ✓ Container redis1                Started                                    185.0s
 ✓ Container redis2                Started                                    184.2s
 ✓ Container redis-cluster-setup   Started                                     74.3s
root@honor13:/home/redis/redis-server#
```

Résultat de la VM backup

```
root@Redis0:~/redis-ha# docker compose up -d

[+] Running 1/1
 ✓ Volume "haproxy-keepalived-data"  Cre...                                   1.2s
WARN[0008] Found orphan containers ([keepalived redis3 redis1 redis-replica-3 redis-
replica-2 redis-replica-1 redis2]) for this project. If you removed or renamed this
[+] Running 2/2 compose file, you can run this command with the --remove-orphans fla
 ✓ Volume "haproxy-keepalived-data"  Cre...                                   1.2s
 ✓ Container haproxy-backup          Started                                 18.0s
root@Redis0:~/redis-ha#
root@Redis0:~/redis-ha#
```

Étape suivante : Tests unitaires

✅ Phase 2 – Tests HAProxy & Keepalived (VIP)

Verification de la VIP ✅

```
root@honor13:/home/redis/redis-server# ip a | grep 10.0.50.4
    inet 10.0.50.4/32 scope global eth0
root@honor13:/home/redis/redis-server#
```

Test de connectivité via le client Redis avec la VIP sur les deux HAProxy ✅ etConnexion au serveur Redis via le client Redis sur les deux HAProxy ✅

```
root@honor13:/home/redis/redis-server# redis-cli -h 10.0.50.4 -p 6390 ping
PONG
root@honor13:/home/redis/redis-server# redis-cli -h 10.0.50.4 -p 6390
10.0.50.4:6390>
```

Opérations Redis via la VIP ✅

```
10.0.50.4:6390> redis-cli -h 10.0.25.168 -p 6379 SET test1 "hello-vip"
(error) ERR unknown command 'redis-cli', with args beginning with: '-h' '10.0.25.168
' '-p' '6379' 'SET' 'test1' 'hello-vip'
10.0.50.4:6390> SET test1 "hello-vip"
OK
10.0.50.4:6390> GET test1
"hello-vip"
```

Vérification des logs HAProxy / Keepalived ✅ et test de bascule

Test de bascule haute disponibilité (HA) ✅.

Le logs haproxy deviennent

```
root@honor13:/home/redis/redis-server# docker stop haproxy-master
```

```
May 16 09:07:21 honor13 Keepalived_vrrp[4096]: (VI_1) received a
!
May 16 09:07:22 honor13 Keepalived_vrrp[4096]: (VI_1) Entering M
May 16 09:07:23 honor13 Keepalived_vrrp[4096]: (VI_1) received a
```

Les logs haproxy Master deviennent

```
root@honor13:/home/redis/redis-server# docker logs -f haproxy-master
2025-05-19T16:09:45.525262+00:00 honor13 Keepalived[18]: Starting Keepalived v2.3.1 (05/24,2024)
2025-05-19T16:09:45.525558+00:00 honor13 Keepalived[18]: Running on Linux 6.8.12-10-pve #1 SMP PREEMPT_DYNAMIC PMX 6.8.12-10 (2025-04-18T07:39Z) (built for Linux 6.6.0)
2025-05-19T16:09:45.525605+00:00 honor13 Keepalived[18]: Command line: 'keepalived' '-i' 'main'
2025-05-19T16:09:45.526916+00:00 honor13 Keepalived[18]: Configuration file /etc/keepalived/keepalived.conf
```

Résultats de la VMBackup

```
root@Redis0:~/redis-ha# docker logs haproxy-backup | grep STATE
2025-05-19T15:56:06.414021+00:00 Redis0 Keepalived_vrrp[21]: (VI_1) Entering BACKUP
STATE
2025-05-19T15:56:10.223843+00:00 Redis0 Keepalived_vrrp[21]: (VI_1) Entering MASTER
STATE
```